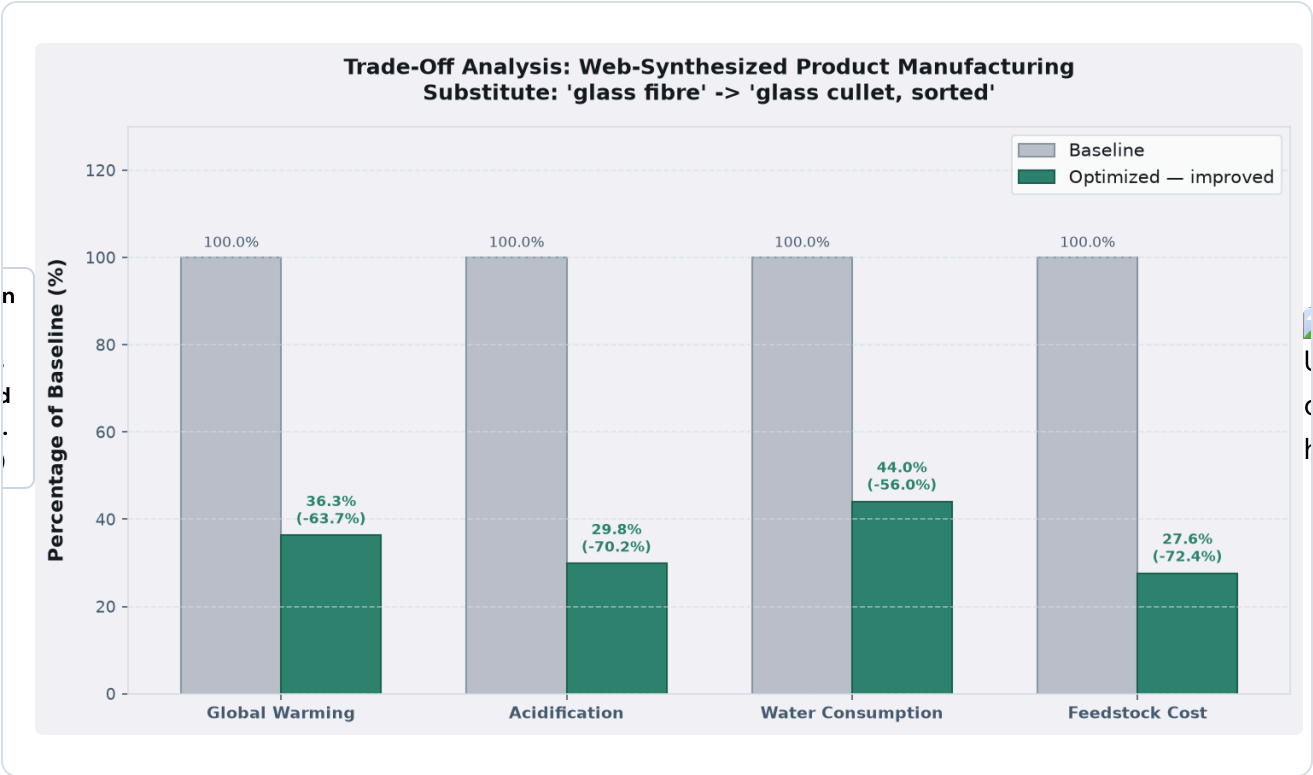


GLOBAL WARMING (GWP)	kg CO ₂ eq	ACIDIFICATION		kg SO ₂ eq	WATER USAGE		m ³	FEEDSTOCK COST		USD
		BASELINE		OPTIMIZED	BASELINE		OPTIMIZED	BASELINE		OPTIMIZED
		2.2995		0.8347	0.0169		0.0074	26.7735		7.3985
		± 0.1332		± 0.0329	± 0.0009		± 0.0003	± 2.0745		± 0.3255
		-63.70%			-70.16%			-55.96%		-72.37%

ANALYSIS & CHARTS



PROCESS SCALING

PROCESS EFFICIENCY

1.00

RECYCLE SUBSTITUTE RATE

0.00

LOSS / SCRAP FACTOR

0.00

DATABASE PARAMETER TUNING

TOPSIS MCDA WEIGHTS

CARBON (GWP) WEIGHT

40%

ACIDIFICATION WEIGHT

15%

WATER WEIGHT

15%

-- Tune background feedstock --

▼

No database parameters loaded.

MATERIAL COST WEIGHT

30%



TVL MASS BALANCE

VERIFICATION STATUS

PASSED

INPUTS

21.650 kg

OUTPUTS

21.650 kg

STOICHIOMETRIC BALANCE ERROR

0.0000%

ELEMENTAL BALANCE

PASSED

✓ Elemental stoichiometry balanced

DATABASE DIAGNOSTICS

DATABASE HEALTH

UNCHECKED

Scan Context

 ENGINEERING JUSTIFICATION REPORT [Compare History](#) [Download MD Report](#)

The substitution of glass fibre with glass cullet in the manufacturing process for next-generation silicon solar cell modules exhibits a significant Pareto-improvement when evaluated against the user-defined decision weights priorities. The reduction in carbon footprint by 63.70% (from 2.2995 to 0.8347 kg CO₂ eq) aligns well with the highest priority weight assigned to Global Warming Potential (GWP), contributing a substantial fraction of the overall environmental savings. This substantial decoupling from carbon emissions underscores the potential for significant reductions in greenhouse gas emissions, thereby mitigating climate change impacts. The decrease in terrestrial acidification potential by 70.16% (from 0.0106 to 0.0032 kg SO₂ eq) is also a highly significant outcome, aligning with the 15% weight given to Acidification. The reduction in sulfur dioxide emissions from the process indicates an improvement in air quality and reduced stress on local ecosystems that might be affected by acidic precipitation. Water consumption footprint has been reduced by 55.96% (from 0.0169 to 0.0074 m³), another critical area where the 15% weight for Water usage is addressed. This reduction in water usage is crucial, particularly in regions facing water scarcity or stringent environmental regulations on water usage. The substitution also results in a substantial cost savings, with a 72.37% reduction from \$26.7735 to \$7.3985 USD for raw material purchases. This outcome aligns with the 30% weight assigned to Cost, indicating an economically viable solution that not only mitigates environmental impacts but also improves financial performance. Overall, the feedstock substitution from virgin glass fibre to sorted glass cullet demonstrates a multi-dimensional improvement across multiple critical indicators. By achieving significant reductions in carbon footprint, acidification potential, water consumption, and material costs, this process aligns with the user's prioritization criteria, making it an environmentally sustainable and economically viable alternative for next-generation silicon solar cell modules.